

Jason Betsargon

Palm Desert, CA | U.S. Citizen | (760) 831-9020 | jasonbetsargon@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Education

Georgia Institute of Technology

Atlanta, GA

M.S. Computer Science, Machine Learning Specialization; GPA: 3.66/4.0

Expected Dec. 2028

– **Relevant Coursework:** Software Development Process, Machine Learning for Trading, Robotics: AI Techniques, Game AI

California State University, San Bernardino

San Bernardino, CA

B.S. Computer Science; GPA: 3.39/4.0

May 2022

– **Relevant Coursework:** Software Engineering, Operating Systems, Database Systems, Machine Learning, Compilers

Technical Skills

Languages: Python, JavaScript, TypeScript, C/C++, SQL, HTML/CSS

Frameworks & Libraries: React, Node.js, Express, Jest, Pytest, Pandas, NumPy, Matplotlib, Bootstrap

Cloud & Tooling: AWS (EC2, S3, IAM), Docker, GitHub Actions, Git, Linux, PostgreSQL

Projects

ML Trading Strategy | *Python, Pandas, NumPy, Matplotlib* | [GitHub](#)

- Built a Random Forest classification learner (1,000 bagged trees, leaf size 10) over five engineered technical indicators to produce daily long/short/flat signals for a single equity.
- Engineered five technical indicators from raw adjusted-close data—Bollinger %B, Momentum, MACD histogram, RSI, and CCI—each with tuned entry thresholds feeding a vote-aggregation signal generator.
- Achieved positive returns over a two-year out-of-sample evaluation while the buy-and-hold benchmark returned **-8.4%**; built a market simulator with configurable commission and market impact to measure strategy degradation as impact increased from 0% to 4% of trade value.

Battleship | *JavaScript, HTML/CSS, Jest* | [GitHub](#) | [Live Demo](#)

- Developed a modular browser-based Battleship game using JavaScript ES6 modules and an event-driven architecture separating game logic from UI rendering.
- Achieved **90%** statement coverage across core modules using Jest and Test-Driven Development with a Red-Green-Refactor workflow.
- Implemented a heuristic enemy AI that switches from random search to targeted hunt mode after a successful hit, reducing the average shots required to sink a fleet versus purely random targeting.

React Resume Builder | *React, JavaScript, HTML/CSS* | [GitHub](#) | [Live Demo](#)

- Built a React single-page application for resume generation with live preview, driven by lifted state and reusable controlled components supporting dynamic form sections and input validation.
- Implemented localStorage persistence with automatic state restoration, and one-click PDF export via react-to-print with print-specific CSS.

Experience

Customer Service Assistant

Palm Desert, CA

Ironsmith, Inc. — architectural metal fabrication, small business

Sept. 2022 – Present

- Hired into customer service; scope expanded to operations, quoting, and order management, owning custom fabrication orders from intake through production scheduling and delivery.
- Serve as the primary internal stakeholder on an **18-month** engagement with a contracted software development team building a custom quote-and-order processing application and company website: gathering requirements from sales, design, and production, specifying feature behavior, and prioritizing the build backlog.
- Review design iterations with the development team in monthly working sessions and validate releases against live order workflows, identifying specification gaps and edge cases before rollout to office staff.
- Built Excel reporting workbooks with visualizations tracking production punctuality and revenue lost to discounts and order errors, replacing ad hoc manual review with recurring reporting used by ownership.

Team Leader, CODE Summer Program

Palm Desert, CA

College of the Desert

Summer 2019 & Summer 2020

- Led programming, electronics, sensor, and cybersecurity workshops for **50+** middle and high school students across a six-week technology program.
- Mentored students through hands-on technical projects, developing programming, problem-solving, and computational thinking skills.

Certifications

- **AWS Certified Cloud Practitioner** — Amazon Web Services, 2024
- **NVIDIA Deep Learning Institute**, 2025 — Fundamentals of Deep Learning; Generative AI with Diffusion Models; Applications of AI for Anomaly Detection